

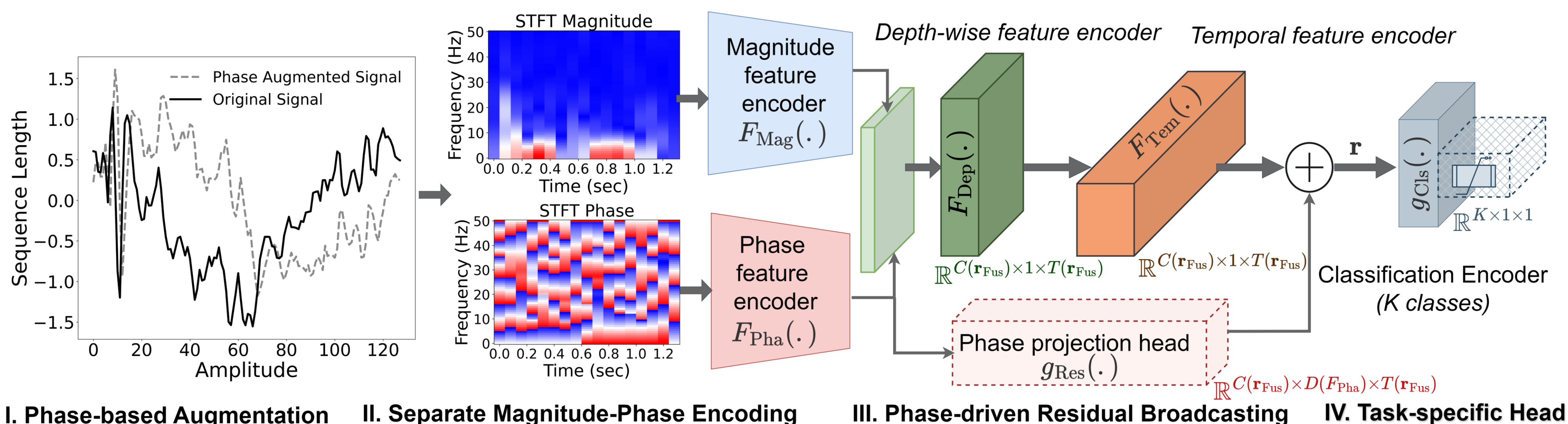


Empowering Real-World Sensing : Algorithms for Imperfect Time-Series

Payal Mohapatra | PhD Candidate, Northwestern University

Generalizable Learning

Domain generalization for nonstationary time-series classification tasks without accessing domain labels for various sensing applications^[2]. [T1, H2]



Phase as a non-stationarity dictionary → Phase-driven learning

- ✓ Hilbert-transform based phase augmentation,
- ✓ Separate magnitude-phase encoding
- ✓ Broadcasted phase-residual connections.

What are the challenges of data-driven sensing time-series analysis?

- T1. Nonstationarity
- T2. Missingness
- T3. Dynamic modalities
- T4. Irregular Sampling

How to handle human-factors in sensing applications?

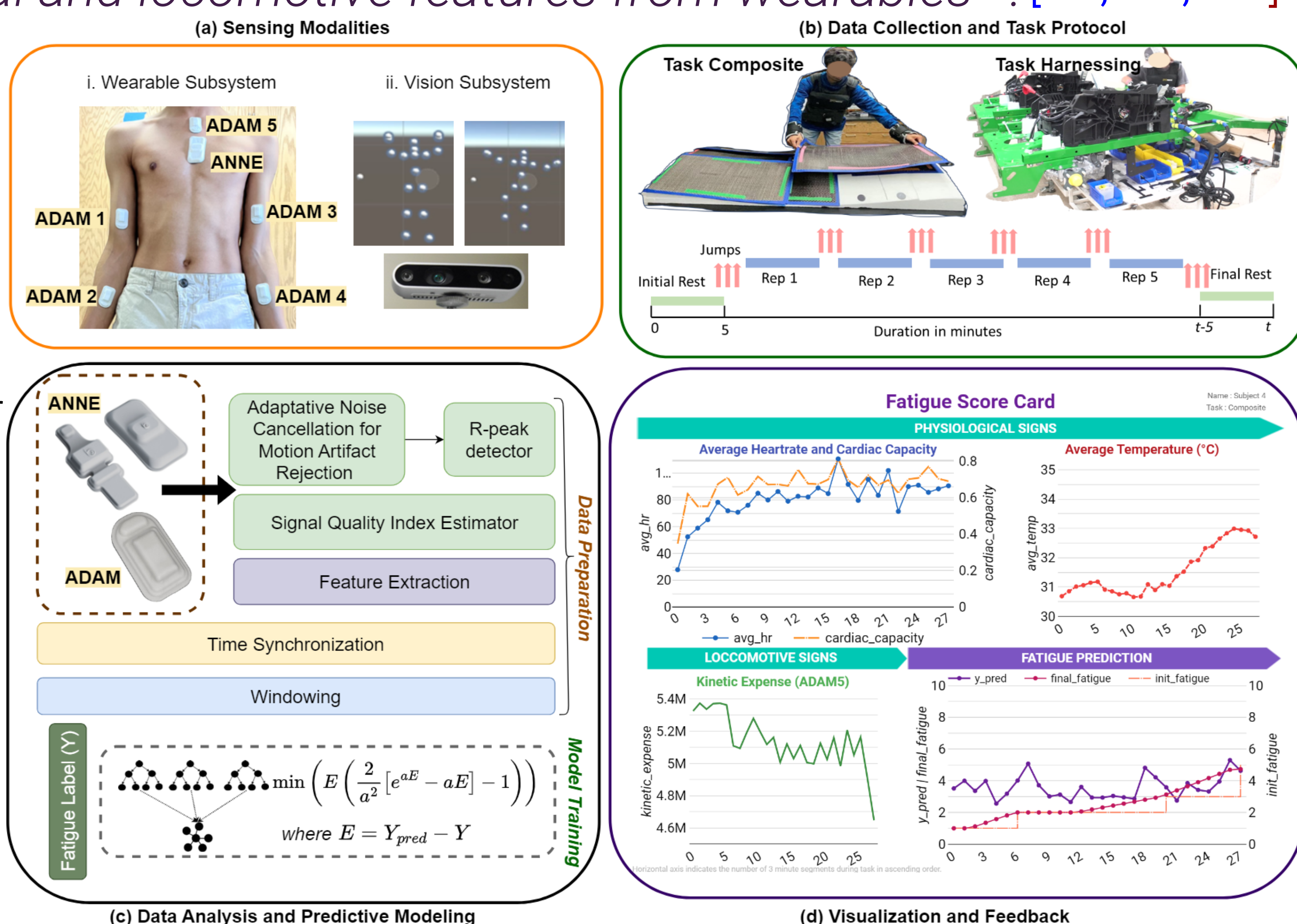
How can we ensure technology is **inclusive and accessible**?

- H1. Subjective Labels
- H2. Subpopulation Distribution Shift
- H3. Minority Users

Physical Fatigue Monitoring

Continuous fatigue prediction in manufacturing workers using physiological and locomotive features from wearables^[1]. [T3, T4, H1]

- Subjective target.
- Real-time.
- Privacy preserving.
- Asymmetric objective for operator-centric design.
- Ensemble tree-based regression.
- Non-visual wearables.

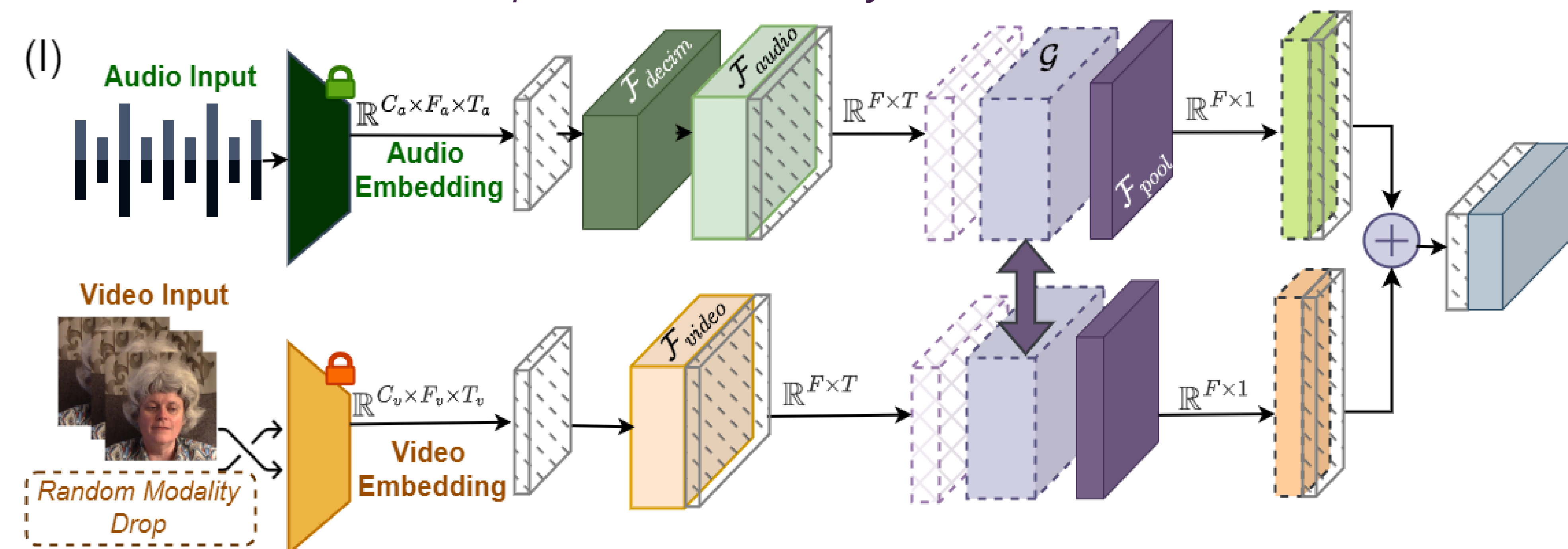


References :

- Mohapatra, P., Aravind, V., Bisram, M., et al. (2024). Wearable network for multilevel physical fatigue prediction in manufacturing workers. PNAS Nexus
- Mohapatra, P., Wang, L., & Zhu, Q. (2024). Phase-driven Domain Generalizable Learning for Nonstationary Time Series. arXiv preprint arXiv:2402.05960
- Mohapatra, P., Pandey, A., et al. (2022). Speech disfluency detection with contextual representation and data distillation, In ACM IASA.
- Mohapatra, P., Islam, B., Islam, M. T., Jiao, R., & Zhu, Q. (2023). Efficient stuttering event detection using siamese networks. In ICASSP 2023.
- Mohapatra, P., Likhite, S., Biswas, S., Islam, B., & Zhu, Q. (2024). Missingness-resilient Video-enhanced Multimodal Disfluency Detection. Interspeech 2024.

Missingness Resilient Learning

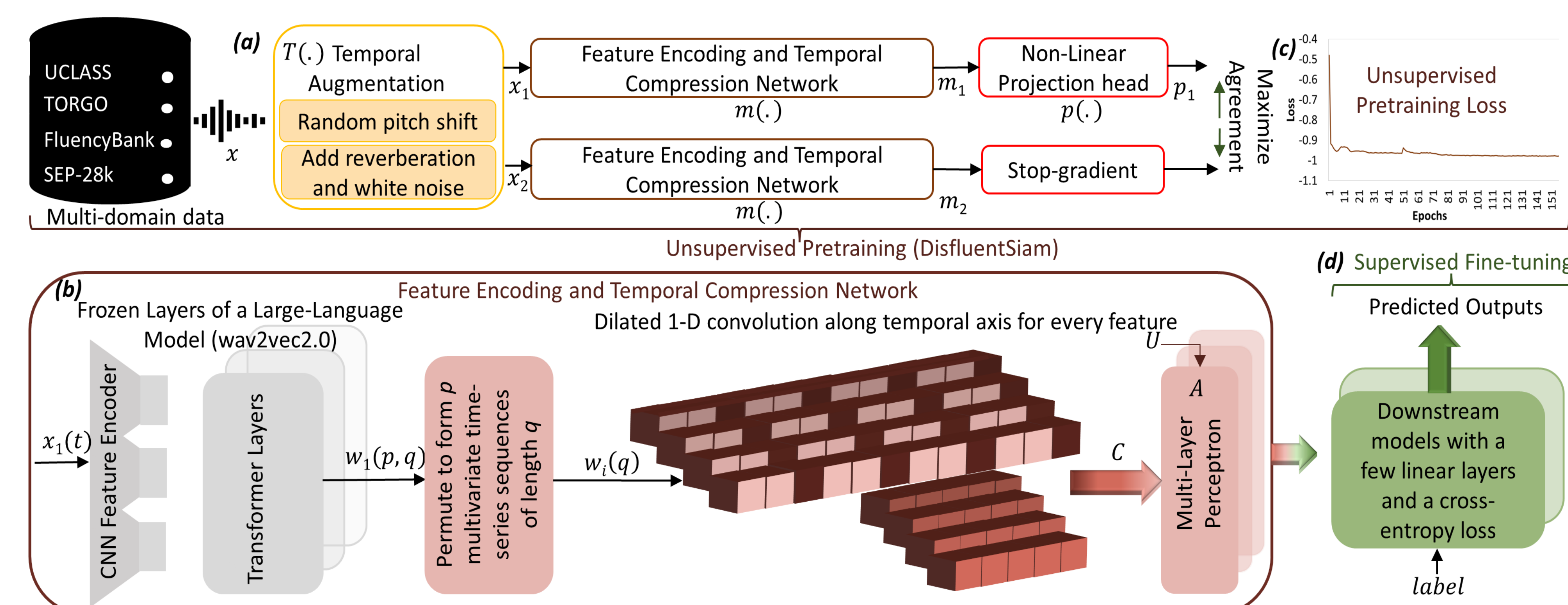
Unified weight-sharing architecture to learn from dynamic audio-visual modalities for speech disfluency tasks^[5]. [T2, T3, H3]



- Scarcity of visual data.
- Complete modality training → inference capability with only audio.
- Weight-sharing encoder design.
- Modality dropout and learnable late-fusion.

Speech Disfluency Detection

1% of the global population stutters and voice assisted technologies are inaccessible to them. [H3]



- Scarcity of data.
- No Labels or Noisy Labels.
- Personalization
- Self-supervised learning^[4].
- Data Distillation^[3].
- Transfer Learning

